HIGHER ORDER DERIVATIVES

	Function	1st Derivative	2nd Derivative	3rd Derivative	nth Derivative
Lagrange	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(n)}(x)$
Leibniz	y	$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	$\frac{d^ny}{dx^n}$

Example 1: Determine the 2nd derivative of $f(x)=x^4+3x^2-5$.

$$f'(x)=4x^3+6x$$

$$f''(x)=12x^2+6$$
$$=6(2x^2+1)$$

Example 2: Determine $\frac{d^2y}{dx^2}$ given that $y=\frac{3x-5}{2x}$. $\rightarrow y=\frac{3}{2}-\frac{5}{2x}$

$$y=\frac{3}{2}-\frac{5x^{-1}}{2}$$

$$\frac{dy}{dx}=\frac{5}{2x^2}$$
$$=\frac{5x^{-2}}{2}$$

$$\frac{d^2y}{dx^2}=\frac{-5}{x^3}$$

Example 3: Determine $f'''(x)$ of the following functions.

a) $f(x)=\frac{1}{x}$ $\rightarrow x^{-1}$

$$f'(x)=-x^{-2}$$
$$f''(x)=2x^{-3}$$
$$f'''(x)=-6x^{-4}$$
$$=\frac{-6}{x^4}$$

b) $f(x)=4x^7+5x^6-8x^4+23x-\frac{\pi}{3}$

$$f'(x)=28x^6+30x^5-32x^3+23$$
$$f''(x)=168x^5+150x^4-96x^2$$
$$f'''(x)=840x^4+600x^3-192x$$

c) $f(x)=\sqrt{x}$ $\rightarrow x^{\frac{1}{2}}$

$$f'(x)=\frac{1}{2}x^{-\frac{1}{2}}$$
$$f''(x)=\frac{-1}{4}x^{-\frac{3}{2}}$$
$$f'''(x)=\frac{3}{8}x^{-\frac{5}{2}}$$
$$=\frac{3}{8\sqrt{x^5}}$$

Velocity and Acceleration

- **Position**, $s(t)$, is the location of an object at time, t.
- **Velocity**, $v(t)$, is the rate of change of position over time, so $v(t) = s'(t) = \frac{ds}{dt}$.
- **Acceleration**, $a(t)$, is the rate of change of velocity over time, so $a(t) = v'(t) = s''(t)$

or $a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$.

Example 1: The height of a soccer ball above the ground at time, t after it is kicked into the air, is given by the formula

$$h(t) = -5t^2 + 30t + 1$$

where h is the height in metres, t is the time in seconds, and $t \geq 0$.

a) Determine an equation that will calculate the instantaneous rate of change of the height of the ball with respect to time, t.

$v(t) = h'(t) = -10t + 30$

b) Determine the velocity of the ball 0.5 seconds after it is kicked. At what other time will the ball have the same speed?

$v(0.5) = -10(0.5) + 30$

$v(0.5) = 25$ m/s

$v(t) = -25$ m/s.

$-10t + 30 = -25$

$-10t = -55$

$t = 5.5$ seconds

c) At what time will the ball reach its maximum height? What is the maximum height reached by the ball?

Max. height occurs at $v(t) = 0$

$-10t + 30 = 0$

$t = 3$ seconds

$h(3) = -5(3)^2 + 30(3) + 1$

$h(3) = 46$ m.

d) What is the instantaneous velocity of the ball as it hits the ground?

$$h(t) = 0$$

$$t = \frac{-30 \pm \sqrt{900+20}}{-10}$$

inadmissable.

$$t \doteq 6.03 \text{ and } -0.033$$

$$\therefore \; v(6.03) \doteq -10(6.03) + 30$$

$$v(6.03) \doteq -30.3 \text{ m/s}$$

Example 2: A freight train leaves a train station and travels due north on a track. After t hours, the train is $s(t) = 18t^2 - 2t^3, 0 \le t \le 9$ kilometres north of the train station .

a) Determine an expression for the velocity of the train at any time $0 \le t \le 9$.

$$v(t) = s'(t)$$

$$v(t) = 36t - 6t^2, \quad 0 \le t \le 9$$

b) Determine the velocity of the train after 3 hours and after 7 hours.

3 hrs	7 hrs
$v(3) = 36(3) - 6(3)^2$	$v(7) = 36(7) - 6(7)^2$
$v(3) = 108 - 54$	$v(7) = 252 - 294$
$v(3) = 54$ km/hr	$v(7) = -42$ km/hr.

c) Determine the acceleration of the train after 1 hour.

$a(t) = v'(t)$	$a(1) = 36 - 12(1)$
$a(t) = 36 - 12t$	$a(1) = 24 \text{ km/hr}^2$

Additional Questions

1. Given $f(x) = ax^2 + bx + c$ and $f(1) = 8$, $f'(1) = 3$ and $f''(1) = -4$, determine the values of a, b and c.

$* \; f'(x) = 2ax + b$

$* \; f''(x) = 2a \Rightarrow 2a = -4 \Rightarrow \boxed{a = -2}$

$* \; f'(x) = 2ax + b$

$f'(1) = 2(-2)(1) + b = 3$

$-4 + b = 3$

$\boxed{b = 7}$

$* \; f(1) = (-2)(1)^2 + 7(1) + c = 8$

$-2 + 7 + c = 8$

$\boxed{c = 3}$

2. The position of a particle is given by $s(t) = 2t^3 - 39t^2 + 216t, \quad t \geq 0$. Determine the acceleration of the particle at the instant when the velocity is zero.

$v(t) = s'(t) = 6t^2 - 78t + 216$

$a(t) = v'(t) = 12t - 78$

$v(t) = 0$

$6t^2 - 78t + 216 = 0$

$t^2 - 13t + 36 = 0$

$(t-9)(t-4) = 0$

$t = 9$ or $t = 4$

$a(9) = 12(9) - 78$

$* \; a(9) = 30 \text{ m/sec}^2$

$a(4) = 12(4) - 78$

$* \; a(4) = -30 \text{ m/sec}^2$